



NUCLEAR PHYSICS — FULL MIND MAP

नाभिकीय भौतिकी · Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

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1. Nuclear Discoveries, Particles & Radiation Detection

- Source matching me Meson—Hideki Yukawa, Positron—C.D. Anderson/U.F. Hess aur stellar-energy theory—H.A. Bethe diye gaye hain; transuranic-element synthesis ko Enrico Fermi se match karna source key ke according incorrect hai. [Q1]
- Nucleus ka splitting process fission hota hai, fusion nahi. Rutherford—Einstein historical assertion aur faster-than-light microwave-photon claim source ke old/current-affairs style statements hain, standard established physics se alag context me padhein. [Q2–Q4]
- Cyclotron charged particles—given options me proton—ko accelerate karta hai; neutral neutron ko directly accelerate nahi karta. Gamma-emitting isotope detect karne ke liye scintillation counter useful radiation detector hai. [Q5–Q6]

⚠ EXAM TRAP — Q1, Q3 & Q4 — Source-specific History/Claims

- ▶ Standard history me Yukawa meson theory/prediction se strongly associated hain; experimental discovery ko later cosmic-ray work se joda jata hai.
- ▶ Q3 ka Rutherford—Einstein wording source-key specific historical assertion hai; ise modern physics principle ke roop me generalize na karein.
- ▶ Q4 me faster-than-light microwave claim established violation of relativity nahi maana jata; exam source wording ko claim ke roop me hi yaad rakhein.

2. Fusion, Fission, Chain Reaction & Nuclear Reactors

- Sun aur normal stars ki primary energy nuclear fusion se aati hai: hydrogen nuclei fuse hoke helium banate hain aur mass-defect energy release hoti hai. Star formation/early stages me gravitational contraction bhi energy contribute kar sakta hai. [Q7–Q11]
- Self-sustained fission chain reaction me har fission se multiple neutrons nikalte hain; moderator (e.g., graphite in source) fast neutrons ko slow karta hai. Har neutron necessarily next fission nahi karata, isliye control zaroori hai. [Q12]
- Nuclear reactor ek controlled self-sustained chain-reaction device/atomic pile hai jo fission heat ko useful energy me convert karta hai; source option me “molecular furnace” synonym accepted hai. [Q13]
- Stars me energy generation fusion se associated hai, fission se nahi. Fusion reactors light nuclei ko join karte hain, typically Tokamak-type magnetic confinement aur extremely high temperature demand karte hain. [Q14–Q15]
- Atom bomb aur fission reactor dono chain reaction use kar sakte hain, lekin reactor me reaction controlled hoti hai; bomb me rapid uncontrolled release hota hai. Radioactivity nuclear property hai, jabki hydrogen bomb ka main principle fusion hai. [Q16–Q17]

3. Thorium, Uranium, Monazite & Nuclear Fuels

- Monazite rare-earth mineral hai aur thorium ka major source hai. Source Q18 me statements 1, 2 & 4 accepted hain; India ke coastal placer deposits thorium-resource base ka important part hain. [Q18]
- India ke context me thorium abundant strategic nuclear resource maana jata hai. Source questions thorium ko large reserve/important future fuel aur uranium ke comparison me abundance, energy-utilization potential aur waste advantage se link karte hain. [Q19–Q21]
- India me electricity generation ka major share historically coal-based raha; power reactors ko 60% enriched uranium ki requirement nahi hoti—civil reactors usually much lower enrichment ya natural uranium designs use karte hain. [Q22]

- Nuclear-fuel family me uranium, plutonium aur thorium major names hain. Cadmium neutron absorber/control-rod material hai; chromium, helium, calcium aur lead given exam options me nuclear fuel nahi hain. [Q23–Q27]

⚠ **EXAM TRAP — Q18, Q21 & Q25 — Context Matters**

- ▶ Monazite processing/export rule regulatory context aur question-year ke saath padhein; source key ko static universal policy na samjhein.
- ▶ Thorium ka “less harmful waste” advantage fuel-cycle design par depend karta hai; source ne statement-set me ise correct maana hai.
- ▶ Q25 me “Radium” ko fuel-element side par chhoda gaya hai, lekin practical reactor fuel ke standard examples U/Pu aur fertile Th hain; option-based GK ko concept se alag rakhein.

4. Atomic/Hydrogen Bombs, Radioactivity & Half-Life

- Atomic bomb ka principle uncontrolled nuclear fission hai; classical fissile material enriched uranium (especially U-235) ya plutonium ho sakta hai. [Q28–Q29]
- Hydrogen/thermonuclear bomb uncontrolled nuclear fusion par based hota hai—deuterium/tritium jaise light isotopes fuse karte hain. Edward Teller ko commonly “father of the hydrogen bomb” kaha jata hai. [Q30–Q32]
- Enriched uranium me natural uranium ke comparison me U-235 proportion artificially increase kiya jata hai. Uranium decay series ka stable end-product lead hai. [Q33–Q34]
- Half-life rule: remaining fraction = $(1/2)^n$. 4-month half-life me $3/4$ decay ke liye 2 half-lives = 8 months; 5-year half-life ke 20 years baad $1/16$ remains; 10-day half-life ke 20 days baad $3/4$ decayed. [Q35–Q37]
- Source Q38 me Zirconium non-radioactive option hai; tritium, astatine aur francium radioactive examples hain. Power generation ke liye fissile U-235 central isotope hai, while uranium metal broadly nuclear-power fuel resource hai. [Q38–Q40]
- Radioactive nuclei spontaneously alpha (α), beta (β) aur gamma (γ) radiation emit kar sakte hain; source question “all of the above” accept karta hai. [Q41]

⚠ **EXAM TRAP — Q38 — Radioactivity Shortcut**

- ▶ “Atomic number $>80 \Rightarrow$ radioactive” ek rough/unsafe shortcut hai, exact rule nahi. Exam me specific isotope/element stability ko priority dein.
- ▶ U-235 fissile fuel isotope hai; U-238 largely fertile component hai aur natural uranium ka major fraction hota hai.

5. Indian Nuclear Research, Kalpakkam & DAE Institutions

- Control rods excess neutrons absorb karke chain reaction regulate karte hain; unke bina reactor reaction out of control ja sakti hai. Common absorber materials me boron/cadmium families aati hain. [Q42]
- India ka first commercial atomic power station Tarapur Atomic Power Station (Maharashtra) hai. Dhruva, Purnima aur CIRUS Indian research reactors/BARC ecosystem se associated hain. [Q43–Q46]
- Indian PHWR fleet natural uranium use kar sakti hai; source option Q47 U-238 ko choose karta hai because natural uranium ka bulk U-238 hota hai, though sustained fission ke liye fissile U-235 fraction crucial hai. [Q47]
- Shivanasamudra hydel power ke liye famous hai; Kalpakkam nuclear energy hub hai. IGCAR Kalpakkam, Tamil Nadu me located hai. [Q48–Q50]
- Kalpakkam FBTR plutonium–uranium mixed carbide fuel system se associated hai aur liquid sodium coolant use karta hai. Source Q53 historical commission key me statements 1 & 3 accepted hain, so FBTR vs PFBR wording ko carefully distinguish karein. [Q51–Q53]
- Harish-Chandra Research Institute Prayagraj me hai, Chennai nahi. Atomic Minerals Directorate, Heavy Water Board, Indian Rare Earths/IREL aur Uranium Corporation source ke according DAE framework se linked organizations hain. [Q54–Q55]
- Heavy Water Board ka nuclear-program role D_2O production/supply hai. Source Q56 Trombay ko answer deta hai; actual heavy-water production network multiple Indian plant locations se historically associated raha hai. [Q56]

⚠ **EXAM TRAP — Q45, Q47, Q53 & Q56 — Time/Source-sensitive**

- ▶ Q45 “present BARC director” source ne itself outdated mark kiya hai; person-name ko question-year ke bina static fact na banayein.
- ▶ Q47 me natural uranium $\approx$ mostly U-238, but U-235 hi fissile component hai—option wording aur physical role alag rakhein.
- ▶ Q53 FBTR/PFBR statements mixed wording use karte hain; source answer key ko historical context me preserve karein.
- ▶ Q56 Trombay answer source-specific hai; Heavy Water Board ke production facilities multiple sites par rahe hain.

6. Nuclear Power Stations, Locations & Capacity

- Key station–state mapping: Kalpakkam–Tamil Nadu, Narora–Uttar Pradesh, Kakrapar–Gujarat, Trombay–Maharashtra. Rajasthan Atomic Power Station Rawatbhata (Rajasthan) me hai, Udaipur nahi; Kaiga Karnataka me hai, Tamil Nadu nahi. [Q57–Q59]
- Given locations me Narora aur Kalpakkam atomic power stations hain; Ghatshila/Nangal ko nuclear power station na samjhein. Narora source question me Seismic Zone IV se associated hai. [Q60–Q61]
- Source-era “recently active” plant Q62 me Kaiga hai. Kudankulam Tamil Nadu me Russian collaboration ke saath planned six-reactor site ke roop me source me diya gaya hai. [Q62–Q63]
- Installed-capacity questions Q71–Q72 source/year context me Kudankulam ko highest answer dete hain. Capacity dynamic data hai, isliye exam year ko always note karein. [Q71–Q72]
- India ke total electricity/energy mix me nuclear share wala Q73 source ne outdated/disputed mark kiya hai; percentage ko current-static GK ke roop me memorize na karein. [Q73]



EXAM TRAP — Q62 & Q71–Q73 — Date-dependent Data

- ▶ “Recently active”, installed capacity aur percentage share time-sensitive facts hain.
- ▶ Source key ko us exam-year snapshot ke roop me padhein; later expansions/commissioning se rankings change ho sakti hain.

7. Moderators, Heavy Water, PHWR & Breeder Reactors

- Heavy water (D_2O) neutron moderator ke roop me fast neutrons ko slow karta hai. Moderator ka purpose neutron energy reduce karke further fission probability ko suitable banana hai. [Q64]
- Different reactor designs me ordinary/light water, heavy water aur graphite moderator ho sakte hain. Source Q65–Q66 me multiple correct answers mark kiye gaye hain; graphite ka role moderator hai. [Q65–Q67]
- Fast breeder reactor electricity fission se produce karta hai, but defining feature ye hai ki fertile material se consumed fissile fuel se zyada new fissile material “breed” kar sakta hai. [Q68–Q69]
- NPCIL PHWR export-capacity Q70 source ne disputed mark kiya hai because cited 220 MWe + 540 MWe pair options me absent tha; exact option ko force na karein. [Q70]

8. Manhattan Project, Pokhran Tests & India’s Nuclear Doctrine

- Manhattan Project World War II ka secret Allied/U.S.-led R&D program tha jisne first atomic weapons develop kiye. [Q74]
- India ka 1974 Pokhran-I test “Smiling Buddha” code name se associated hai. Pokhran-II series 11 May 1998 se start hui aur “Operation Shakti” naam se jani gayi; Pokhran Rajasthan me hai. [Q75–Q78]
- Source Q79 India nuclear doctrine ke pillars me No First Use, unilateral test moratorium aur minimum credible deterrence ko include karta hai. [Q79]

9. CERN, ITER, IAEA, NPT, NSG & International Cooperation

- Nuclear Security Summits UN ke formal aegis me periodic summit-series nahi the; International Panel on Fissile Materials IAEA organ nahi, independent expert initiative hai. [Q80]
- CERN world’s leading/large particle-physics laboratory ecosystem hai; acronym historical provisional council name se aaya. Dono facts true ho sakte hain but acronym-origin laboratory-size ka causal explanation nahi. [Q81]
- ITER international fusion project Southern France (Cadarache) me hai. Success se participating India ko fusion-reactor technology/experience develop karne ka major advantage mil sakta hai. [Q82–Q83]
- RTG miniature fission reactor nahi hota; radioactive decay heat ko thermoelectric conversion se electricity me badalta hai, spacecraft systems me long-duration power ke liye use hota hai. Source Q84 statements 2 & 3 accept karta hai. [Q84]
- Source Q85 historical Indian nuclear-policy assertion me international inspection reluctance ko true aur “India will not produce any nuclear bomb” ko false maanta hai. [Q85]
- Treaty-year matching me SALT-II ko 1967 se match karna incorrect hai; source explanation 1979 signing context batata hai. NPT ko India discriminatory framework maankar sign nahi karta; NPT-recognized nuclear-weapon states listed set me China & France hain. [Q86–Q88]

- NSG membership advanced nuclear trade/technology access facilitate kar sakti hai, but NPT membership automatic nahi hoti. Source Q90 me Iran non-member option hai; 2008 NSG waiver ke baad France first civil-nuclear cooperation agreement signer diya gaya hai. [Q89–Q91]
- India–IAEA Additional Protocol designated civilian nuclear facilities ko safeguards/inspection framework me laata hai. Source Q93 safeguarded vs unsafeguarded reactors ko imported vs domestic fuel-supply separation se link karta hai. [Q92–Q93]
- Source Q94 ICONS-2024 + Counter Nuclear Smuggling Team statements dono ko correct key karta hai; Q95 India ke nuclear-field cooperation “Action Plan” ko Russia ke saath match karta hai. [Q94–Q95]

EXAM TRAP — Q84, Q90, Q94 & International Affairs

- ▶ RTG radioactive decay heat source hai—active chain-reaction fission reactor nahi.
- ▶ NSG membership/status aur conference-date/current-policy items time-sensitive ho sakte hain; source-year context preserve karein.
- ▶ Q94 ko source-key statement ke roop me note karein; date-specific international-event facts ko fresh exam me re-verify karna best practice hai.

✓ Source coverage: Q1–Q95